

HTML Forms: Collecting User Input

Introduction to Forms

HTML forms are used to collect user input and send it to a server for processing. Forms are essential for:

- User registration and login
- Search functionality
- Surveys and feedback
- Online purchases and orders

The `<form>` Element

The `<form>` tag defines a form and contains all input elements.

Syntax:

```
html
<form action="process.php" method="post">
  <!-- Input fields go here -->
</form>
```

Key Attributes:

Attribute	Description
<code>action</code>	Specifies the URL where the form data will be sent
<code>method</code>	Specifies how data is sent (<code>get</code> or <code>post</code>)
<code>name</code>	Provides a name for the form

Common Form Input Types

Input Type	Description	Example
text	Single-line text input	<code><input type="text" name="username"></code>
password	Password field (text is hidden)	<code><input type="password" name="pass"></code>
email	Email address input	<code><input type="email" name="email"></code>
number	Numeric input	<code><input type="number" name="age"></code>
radio	Radio button (select one option)	<code><input type="radio" name="gender" value="male"></code>
checkbox	Checkbox (select multiple options)	<code><input type="checkbox" name="hobby" value="reading"></code>
submit	Submit button	<code><input type="submit" value="Submit"></code>
reset	Reset button	<code><input type="reset" value="Clear"></code>
textarea	Multi-line text input	<code><textarea name="message"></textarea></code>
select	Dropdown list	<code><select name="country"><option>Philippines</option></select></code>
date	Date picker	<code><input type="date" name="birthdate"></code>
file	File upload	<code><input type="file" name="upload"></code>

Complete Form Example

html

```
<!DOCTYPE html>
<html>
  <head>
    <title>Registration Form</title>
  </head>
  <body>
    <h1>Student Registration</h1>
    <form action="register.php" method="post">
      <label for="fullname">Full Name:</label>
      <input type="text" id="fullname" name="fullname" required>
      <br>

      <label for="email">Email Address:</label>
      <input type="email" id="email" name="email" required>
      <br>

      <label for="password">Password:</label>
      <input type="password" id="password" name="password">
      <br>

      <label>Gender:</label>
      <input type="radio" name="gender" value="male"> Male
      <input type="radio" name="gender" value="female"> Female
      <br>

      <label for="country">Country:</label>
      <select id="country" name="country">
        <option value="ph">Philippines</option>
        <option value="us">United States</option>
        <option value="jp">Japan</option>
      </select>
      <br>

      <label>Interests:</label>
      <input type="checkbox" name="interests" value="coding"> Coding
      <input type="checkbox" name="interests" value="gaming"> Gaming
```

```
<input type="checkbox" name="interests" value="reading"> Reading
<br>

<label for="message">Message:</label>
<textarea id="message" name="message" rows="4" cols="50"></textarea>
<br>

<input type="submit" value="Register">
<input type="reset" value="Clear">
</form>
</body>
</html>
```

The `<label>` Element

The `<label>` tag provides a description for form inputs. It improves accessibility and usability. When the user clicks on a label, the associated input field becomes focused.

Using the `for` attribute:

```
html
<label for="username">Username:</label>
<input type="text" id="username" name="username">
```

The `for` attribute must match the `id` of the input element.

Part 4: Introduction to Cascading Style Sheets (CSS)

What Is CSS?

CSS (Cascading Style Sheets) is a language used to describe the presentation of a web page. While HTML provides the structure and content, CSS controls how that content looks — including colors, fonts, layout, spacing, and responsiveness.

Think of it this way:

- **HTML** = The structure of a house (walls, rooms, doors)
- **CSS** = The paint, wallpaper, furniture, and decorations

CSS Syntax

A CSS rule consists of a **selector** and a **declaration block**.

Syntax:

```
css
selector {
  property: value;
}
```

Example:

```
css
p {
  color: blue;
  font-size: 16px;
}
```

Explanation:

- `p` – Selector (selects all paragraph elements)
- `color: blue;` – Declaration (property = color, value = blue)
- `font-size: 16px;` – Declaration (property = font-size, value = 16px)

Parts of a CSS Rule

Part	Description	Example
Selector	Specifies which HTML elements to style	<code>p</code> , <code>h1</code> , <code>.class</code> , <code>#id</code>
Property	The style attribute to change	<code>color</code> , <code>font-size</code> , <code>margin</code>
Value	The specific setting for the property	<code>blue</code> , <code>16px</code> , <code>10px</code>

Types of CSS

Type	Description	Example
Inline CSS	Applied directly to an HTML element using the <code>style</code> attribute	<code><p style="color: red;"></code>
Internal CSS	Defined within the <code><style></code> tag in the <code><head></code> section	<code><style> p { color: red; } </style></code>
External CSS	Defined in a separate <code>.css</code> file and linked to the HTML document	<code><link rel="stylesheet" href="style.css"></code>

Inline CSS

Inline styles are applied directly to an element using the `style` attribute. This method is useful for quick, one-time styling but is not recommended for large projects.

Example:

```
html
<p style="color: blue; font-size: 20px;">This text is blue and larger.</p>
```

Internal CSS

Internal styles are defined in the `<head>` section using the `<style>` tag. They apply to the entire HTML document.

Example:

```
html
<!DOCTYPE html>
<html>
  <head>
    <title>Internal CSS Example</title>
    <style>
      body {
```

```

        font-family: Arial, sans-serif;
        background-color: #f0f0f0;
    }
    h1 {
        color: navy;
        text-align: center;
    }
    p {
        color: gray;
        font-size: 14px;
    }
</style>
</head>
<body>
    <h1>Welcome to My Page</h1>
    <p>This page uses internal CSS.</p>
</body>
</html>

```

External CSS

External styles are defined in a separate `.css` file and linked to the HTML document using the `<link>` tag. This is the preferred method for styling websites because it separates content from presentation, makes styles reusable across multiple pages, and improves maintainability.

Creating a CSS File (style.css):

```

css
body {
    font-family: Arial, sans-serif;
    background-color: #f0f0f0;
}
h1 {
    color: navy;
    text-align: center;
}
p {
    color: gray;
}

```

```
    font-size: 14px;
}
```

Linking the CSS File (index.html):

```
html
<!DOCTYPE html>
<html>
  <head>
    <title>External CSS Example</title>
    <link rel="stylesheet" href="style.css">
  </head>
  <body>
    <h1>Welcome to My Page</h1>
    <p>This page uses external CSS.</p>
  </body>
</html>
```

CSS Selectors

Selector	Description	Example
Element Selector	Selects all elements of a specific type	<code>p { color: red; }</code>
Class Selector	Selects elements with a specific class attribute	<code>.highlight { background: yellow; }</code>
ID Selector	Selects an element with a specific ID attribute	<code>#header { font-size: 24px; }</code>
Universal Selector	Selects all elements	<code>* { margin: 0; }</code>

Class and ID Selectors

Class Selector (.): Applies to multiple elements. A class can be used on any number of elements.

```
html
<p class="highlight">This text is highlighted.</p>
<p class="highlight">This text is also highlighted.</p>
```

```
css
.highlight {
    background-color: yellow;
}
```

ID Selector (#): Applies to a single, unique element. An ID must be unique within a page.

```
html
<h1 id="main-title">Welcome</h1>
```

```
css
#main-title {
    color: navy;
    text-align: center;
}
```

Key Terms

Term	Definition
Hyperlink	A clickable element that navigates to another page or resource
Anchor Tag (<a>)	The HTML tag used to create hyperlinks
Relative Path	A link to a file within the same website using a partial URL
Frameset	A layout that divides a page into multiple frames
Frame (<frame>)	A single section of a frameset that displays a separate HTML document
Form (<form>)	An HTML element that collects user input
Input Field	A form element where users enter data (e.g., text, radio, checkbox)
CSS	Cascading Style Sheets; a language for styling web pages
Selector	The part of a CSS rule that identifies which elements to style
Property	The style attribute being changed in a CSS rule
Value	The specific setting applied to a CSS property
Inline CSS	CSS applied directly to an element using the <code>style</code> attribute
Internal CSS	CSS defined within the <code><style></code> tag in the HTML <code><head></code>
External CSS	CSS defined in a separate <code>.css</code> file and linked to the HTML document

Review Questions

Write your answers on a separate sheet of paper.

1. Write the HTML code to create a link that opens "<https://www.google.com>" in a new tab.
2. What is the difference between an external link and an internal link? Give an example of each.
3. Write the HTML code to create an email link to "contact@school.edu".
4. What is the purpose of the `target` attribute in a link? What value would you use to open a link in a new tab?
5. Write the HTML code for a frameset that divides the page into two columns (30% and 70%).
6. What are the key differences between frames and modern layout techniques like CSS Flexbox?
7. Write the HTML code for a registration form that includes: full name (text), email (email), password (password), gender (radio buttons), country (dropdown), and a submit button.
8. What is the purpose of the `action` and `method` attributes in a `<form>` element?
9. What is the difference between inline, internal, and external CSS? When would you use each?
10. Write a CSS rule that makes all paragraphs have font-size of 18px and color blue.
11. Write the HTML and CSS code to create a class selector named `.highlight` that applies a yellow background to any element.
12. Create a complete HTML page that includes a navigation bar with links to three different pages, a form with at least four different input types, and an embedded image.

Answer Key

1.

html

```
<a href="https://www.google.com" target="_blank">Google</a>
```

2. An **external link** points to a page on another website (e.g., `<a href="https://www.google.com">Google</a>`). An **internal link** points to a page within the same website (e.g., `<a href="about.html">About Us</a>`).

3.

html

```
<a href="mailto:contact@school.edu">Email Us</a>
```

4. The `target` attribute specifies where to open the linked document. The value `_blank` opens the link in a new tab.

5.

html

```
<frameset cols="30%, 70%">
  <frame src="menu.html">
  <frame src="content.html">
</frameset>
```

6. Frames divide a page into multiple separate HTML documents, while CSS Flexbox and Grid provide modern, flexible layout techniques that are easier to control, more responsive, and better for accessibility. Frames are now considered outdated.

7.

html

```
<form action="register.php" method="post">
  <label for="fullname">Full Name:</label>
  <input type="text" id="fullname" name="fullname" required><br>

  <label for="email">Email:</label>
  <input type="email" id="email" name="email" required><br>

  <label for="password">Password:</label>
  <input type="password" id="password" name="password"><br>

  <label>Gender:</label>
  <input type="radio" name="gender" value="male"> Male
  <input type="radio" name="gender" value="female"> Female<br>

  <label for="country">Country:</label>
```

```

<select id="country" name="country">
  <option value="ph">Philippines</option>
  <option value="us">United States</option>
</select><br>

<input type="submit" value="Register">
</form>

```

8. The `action` attribute specifies the URL where the form data will be sent. The `method` attribute specifies how data is sent (`get` or `post`).

9.

- **Inline CSS** – Applied directly to an element using the `style` attribute (best for quick, one-time changes)
- **Internal CSS** – Defined in the `<head>` section within `<style>` tags (best for single-page styles)
- **External CSS** – Defined in a separate `.css` file and linked (best for large, multi-page websites)

10.

```

css
p {
  font-size: 18px;
  color: blue;
}

```

11.

```

html
<!DOCTYPE html>
<html>
  <head>
    <style>
      .highlight {
        background-color: yellow;
      }
    </style>
  </head>
  <body>

```

```
        <p class="highlight">This text is highlighted.</p>
    </body>
</html>
```

12. Sample complete HTML page:

```
html
<!DOCTYPE html>
<html>
    <head>
        <title>My Website</title>
        <style>
            body { font-family: Arial, sans-serif; }
            nav a { margin: 0 10px; text-decoration: none; color: blue; }
        </style>
    </head>
    <body>
        <nav>
            <a href="index.html">Home</a>
            <a href="about.html">About</a>
            <a href="contact.html">Contact</a>
        </nav>
        <hr>

        <h1>Welcome to My Website</h1>
        

        <h2>Contact Form</h2>
        <form action="submit.php" method="post">
            <label for="name">Name:</label>
            <input type="text" id="name" name="name"><br>

            <label for="email">Email:</label>
            <input type="email" id="email" name="email"><br>

            <label>Preferred Contact:</label>
            <input type="radio" name="contact" value="email"> Email
            <input type="radio" name="contact" value="phone"> Phone<br>
```

```
<label for="message">Message:</label>
<textarea id="message" name="message" rows="4" cols="30"></textarea><br>

<input type="submit" value="Send">
</form>
</body>
</html>
```